Q&A 02 Work and Energy

Tuesday, December 12, 2023 10:29 AM

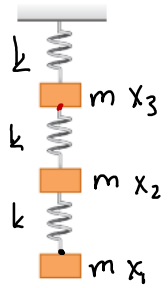
$W = \vec{F}\cdot\vec{s} = \int \vec{F}\cdot d\vec{x}$; $\boxed{\Sigma W = \Delta K}$ $\bar{P} = \frac{W}{t}$; $P = \frac{dW}{dt}$

Joule watt

distance?

6.33 • Three identical 6.40-kg masses are hung by three identical springs, as shown in Fig. E6.33. Each spring has a force constant of 7.80 kN/m and was 12.0 cm long before any masses were attached to it. (a) Draw a free-body diagram of each mass. (b) How long is each spring when hanging as shown? (*Hint:* First isolate only the bottom mass. Then treat the bottom two masses as a system. Finally, treat all three masses as a system.)

Figure **E6.33**



FBD

kx_3, kx_2, mg: $\Sigma F = ma^{\nearrow 0} \Rightarrow kx_3 = kx_2 - 2mg$

$x_3 = 3m$

kx_2, kx_1, mg: $\Sigma F = ma^{\nearrow 0}$

$kx_2 = kx_1 + mg = 2mg \Rightarrow$

kx_1, mg: $a = 0$ $\Sigma F = ma \Rightarrow kx_1 = mg$

$x_1 = \Delta x$; $x_2 = 2\Delta x$ $x_3 = 3\Delta x$

$T_3 = ?$, $T_2 = ?$, $T_1 = ?$

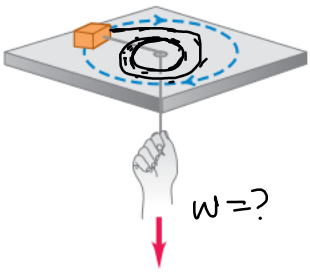
$T_1 = mg \Rightarrow kx_1$

$T_2 = 2mg$

$T_3 = 3mg$

6.75 •• **CP** A small block with a mass of 0.0900 kg is attached to a cord passing through a hole in a frictionless, horizontal surface (Fig. P6.75). The block is originally revolving at a distance of 0.40 m from the hole with a speed of 0.70 m/s. The cord is then pulled from below, shortening the radius of the circle in which the block revolves to 0.10 m. At this new distance, the speed of the block is observed to be 2.80 m/s. (a) What is the tension in the cord in the original situation when the block has speed $v = 0.70$ m/s? (b) What is the tension in the cord in the final situation when the block has speed $v = 2.80$ m/s? (c) How much work was done by the person who pulled on the cord?

Figure **P6.75**



UCM

TOP VIEW

v_1, R_1, a_1

$a_1 = \frac{v_1^2}{R_1}$

$T_1 = m\frac{v_1^2}{R_1} = (0.09)\frac{0.7^2}{0.4}$ N

SIDE VIEW

N, T, mg

v_2, R_2

$\Sigma F_x = ma_x$

$T_2 = m\frac{v_2^2}{R_2} = ($

N, T, mg

$T_2 = 7.056$

$F = const$

$W = \int_{i}^{f} \vec{F}\cdot d\vec{x}$?

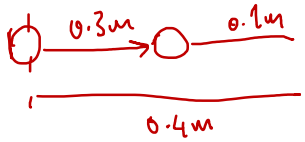
$W = \vec{F}\cdot\vec{x}$

dx, F, f

$\Sigma W = \Delta K \Rightarrow W_N + W_{mg} + W_T = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \frac{1}{2}\,0.09\,(2.8^2 - 0.$

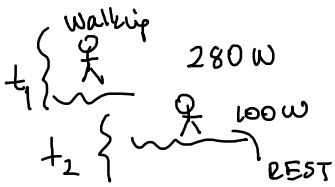
$W_N = 0$, $W_{mg} = 0$

$\rightarrow (7.056 + 0.11025)\,0.2 - 1.075\,T\,($



$F \cdot \Delta x = \left(\frac{\quad}{2} \right)$

6.93 ••• A physics student spends part of her day walking between classes or for recreation, during which time she expends energy at an average rate of 280 W. The remainder of the day she is sitting in class, studying, or resting; during these activities, she expends energy at an average rate of 100 W. If she expends a total of 1.1×10^7 J of energy in a 24-hour day, how much of the day did she spend walking?



$\frac{W}{t} = P$

$t_1 + t_2 = 24\,h$

$t_1 = ?$

$P = \frac{\Sigma W}{\Sigma t} \rightarrow 1.1 \times 10^7\,J \Rightarrow \Sigma W = P_1 t_1 + P_2 t_2$

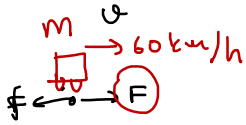
$1.1 \times 10^7\,J = (280\,W)\,t_1 + (100\,W)(86400 - t_1)$

$= 280t + 8.64 \times 10^6 - 100t$

$(11 - 8.64) \times 10^6 = 180t$

$t = 0.01311 \times 10^6\,s = 1.31 \times 10^4\,s = 1$

6.101 •• **Automotive Power II.** (a) If 8.00 hp are required to drive a 1800-kg automobile at 60.0 km/h on a level road, what is the total retarding force due to friction, air resistance, and so on? (b) What power is necessary to drive the car at 60.0 km/h up a 10.0% grade (a hill rising 10.0 m vertically in 100.0 m horizontally)? (c) What power is necessary to drive the car at 60.0 km/h *down* a 1.00% grade? (d) Down what percent grade would the car coast at 60.0 km/h?



Power = 8 hp = 5

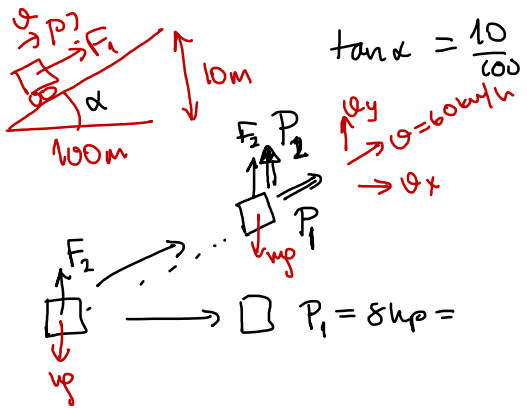
1 hp = 746

$\bar{P} = \frac{W}{t} = 5968\,\frac{J}{s}$

$60\,\frac{km}{hr} =$

$P = \frac{dW}{dt} = \frac{d(\vec{F} \cdot \vec{x})}{dt} = \vec{F} \cdot \frac{d\vec{x}}{dt} = \vec{F} \cdot \vec{v} \Rightarrow$

$F = \frac{P}{v} = \frac{5968\,W}{16.67\,m/s} = 358\,N$



$\tan\alpha = \frac{10}{100} = 0.1$

$\alpha = \tan^{-1}(0.1) = 5.7°$

$\sin\alpha = 0.1$ $\cos\alpha = 0.99$

$P_1 = 8\,hp =$

P_2 is for to raise the car at 10% slope

$P_2 = \vec{F}_2 \cdot \vec{v}_y = \vec{mg} \cdot \vec{v}_y = mg\,v\sin\alpha$

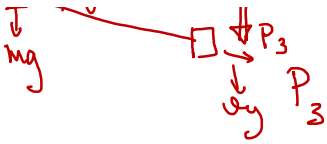
$= (1800)(9.8)($

$= 2.92 \times 10^4\,W$

$\frac{P_2}{746\,W} = \frac{2.92}{746} \times 10^4\,hp = 39.14\,hp$

8 hp + 39.14 hp = total = P

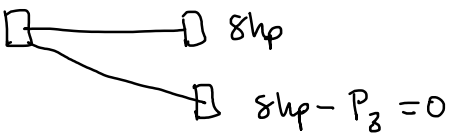
8 hp = P



$P_3 = m\vec{g}\cdot\vec{v}_y = mgv\sin\alpha = 2.92\times10^3 W = 3.91$

$P_{total} = 8hp - 3.91\ hp = \underline{4.09 hp}$

d) yokuş aşağı güç harcamayacağı eğim?



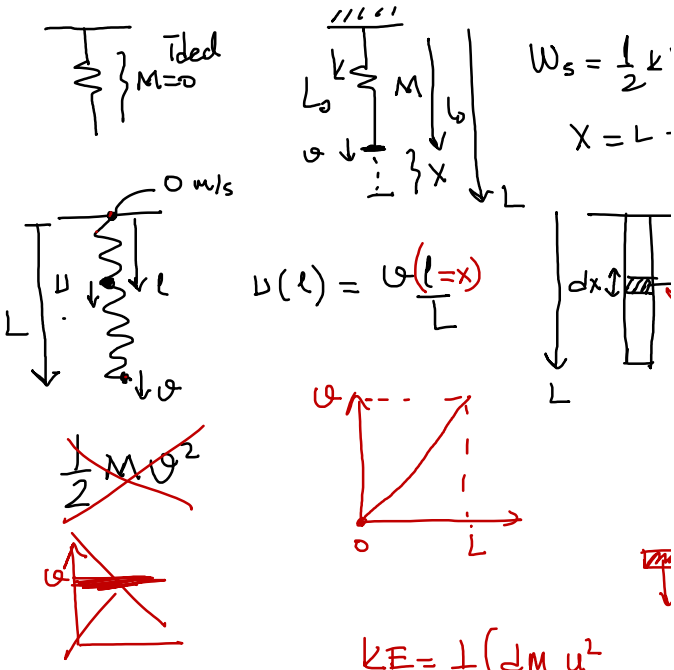
$P_3 = 8hp$

$mgv\sin\beta = 8\times746\ W$ (60 km/h)

$\sin\beta = \frac{8\times}{180}$

$\beta = \sin^{-1}$

6.103 ••• CALC **A Spring with Mass.** We usually ignore the kinetic energy of the moving coils of a spring, but let's try to get a reasonable approximation to this. Consider a spring of mass M, equilibrium length L_0, and spring constant k. The work done to stretch or compress the spring by a distance L is $\frac{1}{2}kX^2$, where $X = L - L_0$. Consider a spring, as described above, that has one end fixed and the other end moving with speed v. Assume that the speed of points along the length of the spring varies linearly with distance l from the fixed end. Assume also that the mass M of the spring is distributed uniformly along the length of the spring. (a) Calculate the kinetic energy of the spring in terms of M and v. (*Hint:* Divide the spring into pieces of length dl; find the speed of each piece in terms of l, v, and L; find the mass of each piece in terms of dl, M, and L; and integrate from 0 to L. The result is *not* $\frac{1}{2}Mv^2$, since not all of the spring moves with the same speed.) In a spring gun, a spring of mass 0.243 kg and force constant 3200 N/m is compressed 2.50 cm from its unstretched length. When the trigger is pulled, the spring pushes horizontally on a 0.053-kg ball. The work done by friction is negligible. Calculate the ball's speed when the spring reaches its uncompressed length (b) ignoring the mass of the spring and (c) including, using the results of part (a), the mass of the spring. (d) In part (c), what is the final kinetic energy of the ball and of the spring?



$KE = \frac{1}{2}\int dM\, u^2$

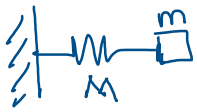
$dM = \frac{M}{L}dx$

$= \frac{1}{2}\int \frac{M}{L}dx$

$= \frac{1}{2}\frac{M}{L^3}v^2\int_0^L x$

$= \frac{1}{2}\frac{M}{L}v^2$

$= \frac{1}{6}Mv^2$



$Energy = \frac{1}{2}mv^2 + \frac{1}{6}Mv^2 = \underbrace{\frac{1}{2}kx^2}_{W_s}$

$X = 0.025 m$ $k = 3200 N/m$ $M = 0.243 kg$ $m = 0.053 kg$

b) $M = 0$ $v = ?$

$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$\sqrt{\frac{(3200)(0.025)^2}{0.053}} = v = 6.14\ m/s$

c) if $M \neq 0$ $\frac{1}{2}kx^2 = \frac{1}{2}mv_1^2 + \frac{1}{6}Mv_1^2$

$\sqrt{\frac{3kx^2}{2m+M}} = v_1 = 3.86\ m/s$

$\sqrt{5m+M}$

$$\frac{1}{2}kx^2 = KE_{ball} + KE_{spring}$$

$$KE_{ball} = \frac{1}{2}(0.053)\,3.86^2 = 0.394\,J$$

$$KE_{spring} = \frac{1}{6}M v_1^2 = \frac{1}{6}(0.243)$$

$$= 0.603\,J$$